

04: Steps Involved to Develop JDBC App (Theory)

You must correctly follow the provided steps to guarantee that your JDBC application can connect to a database, execute SQL commands, process results, and efficiently clean up resources.

1. **Import the Required Package:** To use the JDBC API, you need to import the necessary classes and interfaces from the `java.sql` package.
2. **Load and Register the Driver:** The JDBC driver must be loaded into memory and registered with the `DriverManager` to establish a connection with the database.
3. **Establish the Connection:** Create a connection from the Java application to the database. This requires specifying the database URL, username, and password.
4. **Create the Statement:** To execute SQL queries, you need to create a `Statement` object. This object is used to send SQL commands to the database.
5. **Execute the Query:** The `Statement` object is used to execute SQL queries, such as `SELECT`, `INSERT`, `UPDATE`, or `DELETE`.
6. **Process the Result:** After executing a query, the results are processed.
7. **Close the Connection:** Close the database connection and other resources to free up resources and avoid potential memory leaks.